

Project Overview Document

Team Roles, Implementation Schedule & Key Deliverables

1. Team Roles & Responsibilities

To ensure a high-quality and well-structured delivery, the project has been distributed across three specialized roles, each contributing a distinct layer of the application — from server-side infrastructure to visual design and intelligent feature integration.

Rohan G (241IT064) — Lead Backend Engineer

Rohan serves as the backbone of the project's server-side architecture. He is responsible for engineering all backend logic using **Node.js** and **Express.js**, ensuring a robust, scalable, and maintainable API structure. His scope includes the design and management of the **database architecture**, with flexibility to use either **MongoDB** (for document-based, flexible schemas) or **PostgreSQL** (for relational, structured data), depending on the evolving needs of the platform. A particularly critical aspect of Rohan's work is the implementation of **WebSocket communication via Socket.io**, which powers the real-time responsiveness of the AI chatbot — enabling instant, low-latency interactions between the user and the academic assistant.

Harshith R (241IT032) — UI/UX & Interactive Design

Harshith leads the visual and experiential identity of the platform. He is tasked with building the **interactive 3D/2D virtual workspace** — a digitally immersive study environment rendered using either **Three.js** (for full 3D scenes) or the **Canvas API** (for 2D performance-optimized layouts). A signature feature under his ownership is the **virtual robot guide**, an animated navigation assistant that helps users orient themselves within the platform's various modules. Harshith also ensures the application is visually consistent and device-friendly through **responsive UI layouts** crafted with **Tailwind CSS** or **Bootstrap**, guaranteeing a polished experience across screen sizes.

Pranathi Udaya Kumar (241IT054) — Frontend & AI Integration

Pranathi bridges the gap between design and intelligence. She is responsible for constructing the application's **modular UI component library** using **React.js** or **Vue.js**, ensuring each section of the app is reusable, maintainable, and performant. Her most impactful contributions lie in feature integrations: she connects the **OpenAI API** to power the academic doubt-clearing chatbot — enabling students to ask subject-specific questions and receive intelligent, contextual answers in real time. Additionally, she integrates the **Spotify Web API**, embedding a fully functional music streaming experience directly within the study environment to enhance focus and ambiance.

2. 8-Week Implementation Schedule

The project follows a structured four-phase development plan, progressing from foundational setup to full quality assurance over an eight-week timeline.

Phase 1: Foundation (Weeks 1–2)

The opening phase lays the groundwork for all future development. During **Week 1**, the team focuses on architecture planning and environment configuration — this includes initializing the **Git repository** for version control and setting up the core **Node.js development environment**. A clean, well-organized codebase from day one minimizes technical debt down the line.

Week 2 shifts attention to visual prototyping, where the team designs the **2D/3D study environment**. In this virtual space, real-world study elements — such as books, tables, whiteboards, and shelves — are represented as interactive visual objects, forming the immersive interface students will engage with throughout their sessions.

Phase 2: Module Development (Weeks 3–5)

This phase is the heart of the build, where each core feature module comes to life.

Week 3 introduces the **Study Module** — a productivity-focused system featuring a **task manager with to-do lists**, a **syllabus organizer** to track academic progress, and **local storage persistence** so users never lose their data between sessions.

Week 4 focuses on the **Music Integration** layer. The team implements both **local music playback** (for offline use) and connects to an **external streaming API**, enabling students to curate mood-based playlists directly within their virtual workspace.

Week 5 brings the platform's most intelligent feature to life: the **AI Assistance module**. This involves building the full **chatbot interface** alongside its supporting backend logic, providing students with instant, reliable academic support — from solving equations to explaining complex concepts on demand.

Phase 3: Analytics & Polish (Weeks 6–7)

With core features in place, the team turns to refinement and insight-driven tooling.

Week 6 introduces the **Dashboard & Insights** system — a data visualization layer built with **Chart.js** or **D3.js** that presents meaningful analytics on the user's study patterns, session durations, and task completion rates. This empowers students to understand their own habits and optimize their learning routines.

Week 7 finalizes the **virtual robot guide**, sharpening its contextual awareness and assistant logic so it can intelligently respond to user actions — pointing users toward relevant features, offering tips, and enhancing the overall onboarding experience.

✅ **Phase 4: Quality Assurance (Week 8)**

The final week is dedicated entirely to stability and documentation. The team conducts thorough **API testing via Postman**, validates all integrations end-to-end, addresses any edge-case bugs, and compiles a **comprehensive user guide** — ensuring that both technical reviewers and end users can confidently navigate and evaluate the platform.

3. Key Project Deliverables

At the conclusion of the 8-week development cycle, the project will produce three core deliverables:

💻 **Centralized Study Platform**

A fully integrated **web application** that unifies task management, time tracking, AI-powered academic assistance, and music streaming into a single, cohesive environment — eliminating the need for students to juggle multiple disconnected tools.

🎮 **Interactive Gamified Interface**

A **gamified virtual workspace** designed to reduce cognitive load and sustain student motivation. By presenting study activities within an engaging, visually rich environment, the platform transforms routine studying into a more rewarding and immersive experience.

📱 **Responsive Cross-Device Design**

A fully **responsive application** optimized for both desktop and mobile devices, ensuring students can access their workspace and continue productive study sessions from anywhere — whether at a desk or on the go.

Document prepared for internal project reference and academic submission.